1 · Job position

Site Reliability Engineer (SRE): software engineers who run services and build systems that replace manual operations work

2 · Organization and workload

Google; services SRE runs: Search, Ads, Gmail, Photos

Hundreds of on-call rotations; SRE is separate from product teams

3 · Operating system

Linux kernel underpins much of Google production

Internal Linux distribution; gLinux Rodete (Debian-based) on Googler machines

Borg: cluster OS that runs huge numbers of Linux containers

4 · Hardware and infrastructure

Google-designed datacenters: custom power, cooling, network and compute

Jupiter Clos network inside datacenters; B4 backbone between them

Thousands of machines and disks failed per cluster in a typical year

5 · Software, tools and applications

Blaze/Bazel hermetic builds for C++, Java, Python, Go, JavaScript

Google Front End (GFE) terminates and routes traffic to most services

Practice lab (assumption): Ubuntu VM, Git, Bash/Python, Docker, Prometheus, minikube

6 · Required knowledge and skills

Reliability practice: SLOs, toil limit of 50%, blameless postmortems

OS concepts: scheduling, virtual memory, namespaces and cgroups

Hands-on preparation (assumption): Linux shell, containers, metrics, incident write-up